

Sprint 4 Report

NIFTY100 Financial Intelligence Platform - Dashboard and Valuation Delivery

Sprint Goal

Sprint 4 focused on completing the dashboard integration layer and adding a valuation engine that turns market-cap, financial ratio, cash-flow, company, profit and loss, and sector inputs into decision-ready valuation outputs.

The sprint goal was to deliver a usable dashboard experience with company-level valuation cards, a home-page valuation summary, exportable valuation files, and final QA coverage across dashboard, database, and generated artifacts.

Dashboard Architecture



- ETL loads raw and supporting Excel files into the SQLite warehouse.
- Analytics modules calculate profitability, leverage, CAGR, cash-flow, peer, and valuation metrics.
- Screener ranks companies using configurable quality and value rules.
- Dashboard presents home, profile, peer, trends, sector, capital, and report views.
- Valuation exports summary and flag files consumed by the dashboard.

Eight Dashboard Screens

Screen	Purpose
Home	Executive KPIs, sector distribution, top companies, and valuation summary.
Company Profile	Company metadata, valuation card, revenue/profit chart, ROE/ROCE trend, pros and cons.
Screener	Interactive company filtering and scoring workflows.
Peer Comparison	Peer-group metrics, percentiles, and comparison outputs.
Trend Analysis	Historical metrics and company-level trend exploration.
Sector Analysis	Sector counts, sector-level summaries, and distribution views.

Capital Allocation	Cash-flow pattern classification and allocation labels.
Annual Reports	Company annual-report access and report metadata.

Valuation Engine

The valuation engine was implemented in `src/analytics/valuation.py` with separate functions for loading inputs, calculating FCF Yield, calculating sector median PE, applying PE flags, generating summaries, and exporting results.

Metric	Rule
FCF Yield %	Free Cash Flow / Market Cap * 100
Sector Median PE	Median PE grouped by <code>broad_sector</code>
PE vs Sector	PE / Sector Median PE
Flag: Caution	PE > 1.5 x Sector Median
Flag: Discount	PE < 0.7 x Sector Median
Flag: Fair	All other valid comparisons

Generated `valuation_summary.xlsx` contains 92 companies. Generated `valuation_flags.csv` contains only Discount and Caution companies.

Database Integration

- SQLite warehouse remains the source of operational dashboard data.
- `financial_ratios` table count verified at 1,184 rows.
- `peer_percentiles` table count verified at 7,061 rows.
- Dashboard valuation access uses `get_valuation()` with `@st.cache_data(ttl=600)`.
- Valuation files are read from `output/valuation_summary.xlsx` with CSV fallback support.

Performance

Area	Result
Profile load requirement	Target below 3 seconds
Observed profile loads	0.27s to 0.40s after search submission for QA sample
Caching	Valuation access cached for 600 seconds
Dashboard responsiveness	Desktop and mobile checks completed with no horizontal overflow

QA Results

Check	Result
Full pytest suite	416 passed, 0 failures, 2 warnings
Company QA sample	10 companies across IT, Banking, Energy, Healthcare, and FMCG

Missing values	Displayed as N/A; no raw NaN, None, or null in dashboard checks
Charts	Revenue, Net Profit, ROE, and ROCE legends verified
CSV download	Download Valuation CSV verified successfully
valuation_summary.xlsx	92 rows and 92 unique companies
valuation_flags.csv	44 rows: 30 Discount, 14 Caution, no invalid flags

Deliverables

- src/analytics/valuation.py - valuation calculations and exports.
- output/valuation_summary.xlsx - 92-company valuation summary.
- output/valuation_flags.csv - Discount and Caution companies only.
- src/dashboard/utils/db.py - cached valuation accessor.
- pages/01_home.py - home-page valuation summary and CSV download.
- pages/02_profile.py - company profile valuation card.
- dashboard/dashboard_documentation.md - architecture and screenshot placeholders.
- reports/Sprint4_Report.pdf - sprint report.
- reports/Sprint4_Retrospective.md - sprint retrospective.

Future Enhancements

- Add automated unit tests for valuation formulas and export schemas.
- Add generated screenshots to dashboard documentation.
- Persist valuation summary into SQLite for SQL-native dashboard access.
- Add valuation trend charts for PE, PB, EV/EBITDA, and FCF Yield.
- Extend valuation comparison from sectors to peer groups.
- Add scheduled regeneration of valuation exports after ETL refreshes.

Conclusion

Sprint 4 completed the valuation and dashboard integration layer for the NIFTY100 platform. The project now includes validated valuation calculations, export artifacts, dashboard cards and summaries, CSV download support, and a final QA pass with all automated tests passing.